Carpenter's guide: the studio interior

Everything the carpenter needs to fit out one 20 ft container
as a guest studio, step by step, with every measurement.

Status: revision A2, 2026-10-28, PRELIMINARY. Sizes come from photos until the Day 1 survey; the workshop cuts nothing before revision B.

Everything the carpenter needs to fit out one 20 ft container as a guest studio, step by step, with every measurement. The two studios are the same, so one guide serves both.

Kinyarwanda: soma aya mabwiriza mu Kinyarwanda · Print: this guide as a PDF (not printed yet)

Words used here

- **End A**: the closed end wall, by the bed. **End B**: the cargo-door end, by the bathroom.
- **Back wall**: the long wall with no openings (bed and sofa). **Front wall**: the long wall with the sliding glass door and the small window W1.
- **From the steel**: measured from the bare steel, before any lining. **From the finished wall**: measured from the lining face.
- All sizes are millimetres (mm) unless written otherwise.

The studio at a glance

What	Size (mm)
Container inside, steel to steel	5898 long, 2352 wide, 2393 high
Finished room, lining to lining	5798 long, 2252 wide, 2316 high (13.1 m²)
Wall build-up	50: batten 38 with spray foam between, then plywood 12
Ceiling build-up	60
Floor build-up	17: plywood sheets 12, then oak planks 5 (bathroom: vinyl)
Sliding glass door, from the finished end-A wall	1380 to 4380, head at 2240

What	Size (mm)
Bathroom partition, from the finished end-A wall	starts at 4578, 70 thick

Who	Steps
Carpenter, 2 people	S02, S03, S05 (lining), S06, S07, S11 to S18
Welder	S01 (all hot work)
Electrician and plumber	S04 (first fix), S19 (second fix)
Spray-foam contractor	S05 (foam)
Tiler	S08

Five hold points: stop and check

1. **Survey back.** The Day 1 measurements are in and revision B is issued. Until then the workshop cuts nothing; site steps S01 to S03 can go ahead.
2. **Hot work done.** All welding and grinding (S01) is finished before any foam goes in.
3. **Pressure test passed.** The first fix (S04) is tested at 10 bar and the result written down before the foam.
4. **Foam cured.** The foam has cured for 24 hours and is trimmed flush before the first lining panel.
5. **Wet walls checked.** The bathroom membrane is inspected before the tiler starts.

Two rules that never change:

- Never sand, plane or cut into the container floor: it is treated with pesticide. Cover it with the membrane and screw new sheets on top.
- Set out from the steel, as the tables say, and check the finished sizes against this guide.

Day 1: measure the container

Measure both studio containers before anything else and send the numbers the same day. Mark U1 and U2 on their doors with chalk. The third container is the office; it is not measured here.

Survey sheet, fill the value_mm column · source `study/studio/survey.csv`

unit	item	value_mm	measured_by	date	note
U1	int_length				inside length
U1	int_width				inside width at 1 m height
U1	int_height				floor to roof at mid-width
U1	o1_start				slider opening start from the inside face of end A
U1	o1_end				slider opening end from the inside face of end A
U1	o2_start				narrow door opening start from end A
U1	o2_end				narrow door opening end from end A
U1	diag_diff				floor diagonals difference
U1	level_diff				highest minus lowest corner casting
U2	int_length				inside length
U2	int_width				inside width at 1 m height
U2	int_height				floor to roof at mid-width
U2	o1_start				slider opening start from the inside face of end A
U2	o1_end				slider opening end from the inside face of end A
U2	o2_start				narrow door opening start from end A
U2	o2_end				narrow door opening end from end A
U2	diag_diff				floor diagonals difference
U2	level_diff				highest minus lowest corner casting

1. **int_length**: inside length, steel of end A to steel of end B, 1 m above the floor.
2. **int_width**: inside width, steel to steel, 1 m above the floor, halfway along.
3. **int_height**: floor to roof steel, halfway across.

4. **o1_start** and **o1_end**: where the big opening for the sliding door starts and ends, measured from the steel of end A.
5. **o2_start** and **o2_end**: where the old narrow door starts and ends, from the steel of end A.
6. **diag_diff**: measure both floor diagonals, corner to corner; write the difference.
7. **level_diff**: with a water level, the highest corner casting minus the lowest.
8. Take photos: each wall straight on, the floor, the four corner castings, the sliding-door frame, the old narrow door.

Order of work

Step	Work	Who	Hold point
S01	Survey, level the supports, all welding	Welder and carpenter	1, 2
S02	Membrane and floor sheets	Carpenter	
S03	Battens	Carpenter	
S04	First fix: cables, conduits, pipes	Electrician and plumber	3
S05	Spray foam, then the lining panels	Sprayer, then carpenter	4
S06 to S10	Bathroom: partition, door, wet walls, tray, WC, heater, basin	Carpenter, tiler, plumber	5
S11 to S14	Oak planks, then the kitchen	Carpenter	
S15 to S17	Bed, bedside table, seating	Carpenter	
S18	Trims, caulk, paint, insect screens	Carpenter and painter	
S19	Second fix and certificate	Electrician	
S20 to S27	Roof terrace and stair, later	Welder and carpenter	

S01: survey, level and all the welding

1. Fill the survey sheet (Day 1 above).

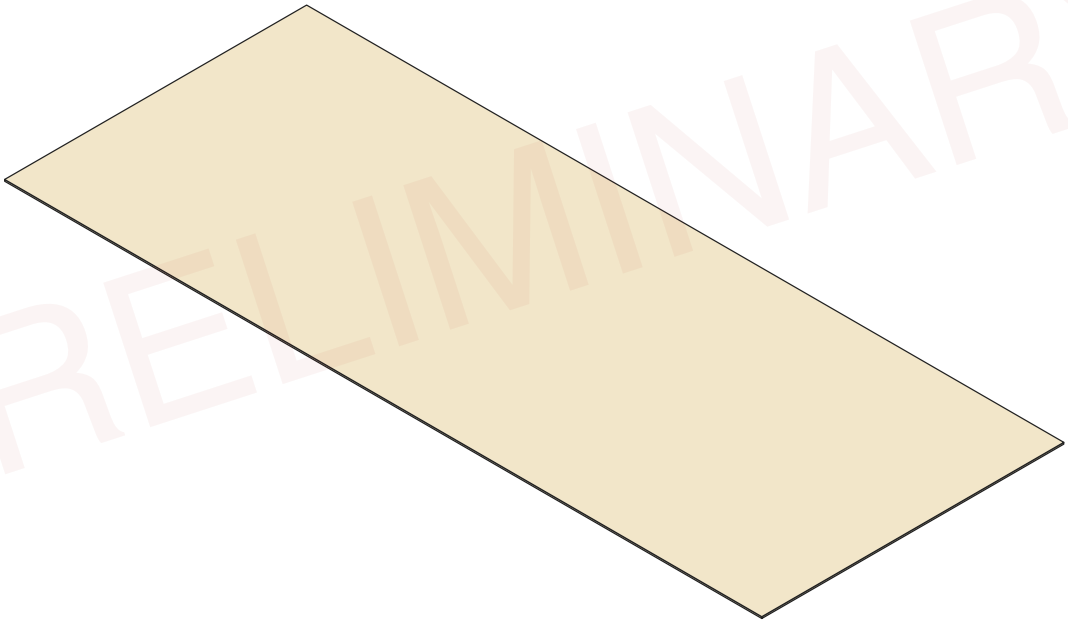
2. Level the four corner castings on concrete pads or steel shims; the corner on the rock needs a pad. Aim for under 5 mm between the highest and lowest corner, and write it in level_diff.
3. Welder: weld the 2 mm steel infill into the old narrow door (4990 to 5510 from the finished end-A wall, up to 1150 high) and fit the louvre window 520 × 890 above it.
4. Welder: seal and lock the end-B cargo doors, and cut the 75 mm sleeve for the AC in the back wall.
5. From S05 on, no welding or grinding on the container: it would burn the foam.

S02: membrane and floor sheets

02

Insulation and lining

Prime the steel; lay the membrane over the treated floor; screw down t



! Never sand the treated container floor. Sheets 2440 x 1220, long side along the box, joint

screw

189x F

Farm container studios	farm.databayt.org/docs/studio
Prime the steel; lay the membrane over the treated floor; screw	**AS-02**
Scale NTS · Rev A2 · 2026-10-28 · mm	PRELIMINARY

PRELIMINARY

AS-02 Membrane and floor sheets · open full size

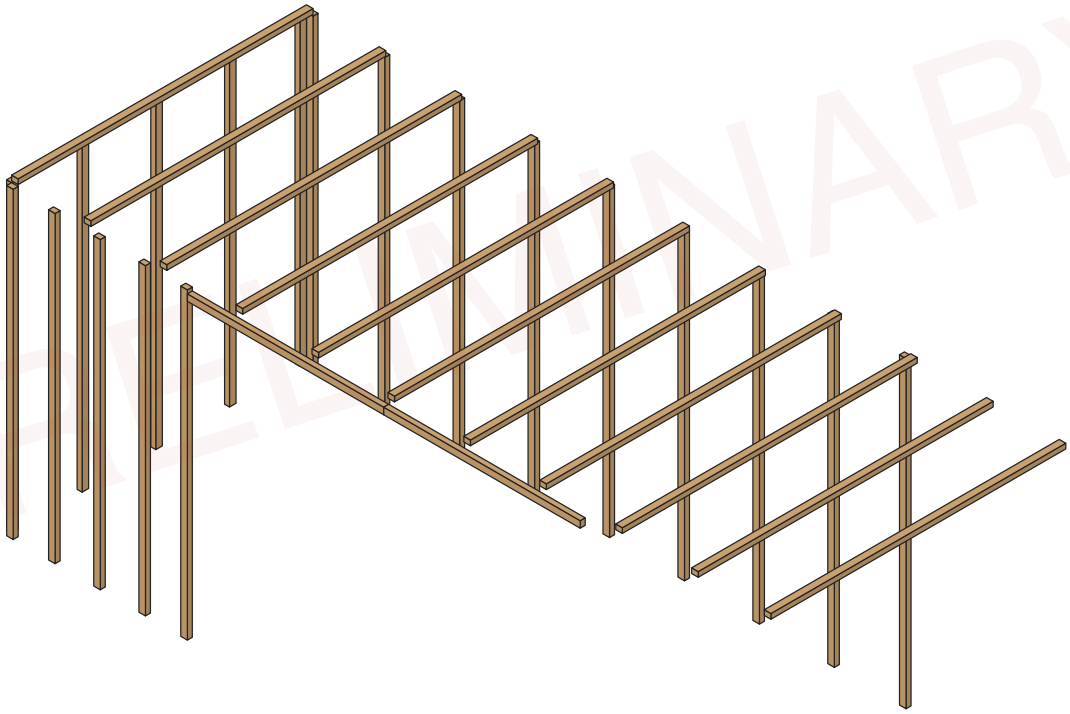
1. Brush rust converter, then zinc primer, on any bare or rusty steel of the walls and ceiling (4 L per studio).
2. Sweep the container floor. Lay the polythene membrane (250 µm) over the whole floor: laps of 150 mm, taped, turned 50 mm up the walls (15.3 m² with the laps).
3. Lay the plywood sheets 12 mm (2440 × 1220), long side along the container, 3 mm gaps between sheets and 5 mm from the steel: 5 sheets per studio.
4. Run 1, along the back wall: a full sheet, another full sheet, then the last one cut to fit (about 1000 long).
5. Run 2, along the front wall, ripped to the width left (about 1120): start with a half sheet (1220 long, from the run-1 offcut), then a full sheet, then the last one cut to fit (about 2220 long). No joint lines up with a joint in run 1.
6. Screw down with wood screws 4 × 30 (bag F) every 300 mm both ways, about 189 per studio. Use 30 mm screws only, so they stay inside the 28 mm container floor.
7. Check: no step at any joint, and the floor flat within 3 mm under a 2 m straightedge.

S03: battens

03

Insulation and lining

Fix the battens: one under every lining-panel joint and edge, infill a



! Winged screws into the steel at both ends and every 600 mm along each batten.

sds
192x L

Farm container studios	farm.databayt.org/docs/studio
Fix the battens: one under every lining-panel joint and edge, in	**AS-03**
Scale NTS · Rev A2 · 2026-10-28 · mm	PRELIMINARY

PRELIMINARY

AS-03 Battens · open full size

1. Use treated battens 38 × 50, standing on the floor sheets: 32 per studio.
2. Put one batten under every lining-panel joint and one between, so no gap is over 600 mm. Mark each centre from the steel, as the set-out table says.
3. Fix each batten to the steel with winged self-drilling screws 5.5 × 50 (bag L): one at each end and one every 600 mm, 192 per studio.
4. Over the sliding door, fix one horizontal batten along the head at 2240 (two lengths of 1500).
5. Check: every batten plumb, and all of them in one plane: a straightedge across three battens touches all three.

Batten set-out, the centre of each batten · download CSV

id	wall	measured	centre_mm	length_mm
K1-BAT-01	back wall	from the steel of end A	75	2316
K1-BAT-02	back wall	from the steel of end A	622	2316
K1-BAT-03	back wall	from the steel of end A	1194	2316
K1-BAT-04	back wall	from the steel of end A	1767	2316
K1-BAT-05	back wall	from the steel of end A	2339	2316
K1-BAT-06	back wall	from the steel of end A	2911	2316
K1-BAT-07	back wall	from the steel of end A	3484	2316
K1-BAT-08	back wall	from the steel of end A	4056	2316
K1-BAT-09	back wall	from the steel of end A	4603	2316
K1-BAT-10	end A	from the steel of the back wall	75	2316
K1-BAT-11	end A	from the steel of the back wall	613	2316
K1-BAT-12	end A	from the steel of the back wall	1176	2316
K1-BAT-13	end A	from the steel of the back wall	1739	2316
K1-BAT-14	end A	from the steel of the back wall	2277	2316
K1-BAT-15	front wall	from the steel of end A	75	2316
K1-BAT-16	front wall	from the steel of end A	395	2316
K1-BAT-17	front wall	from the steel of end A	740	2316
K1-BAT-18	front wall	from the steel of end A	1085	2316

id	wall	measured	centre_mm	length_mm
K1-BAT-19	front wall	from the steel of end A	1405	2316
K1-BAT-20	slider head	horizontal, from the steel of end A	1430-2930	1500
K1-BAT-21	slider head	horizontal, from the steel of end A	2930-4430	1500
K1-BAT-22	ceiling	from the steel of end A	75	2252
K1-BAT-23	ceiling	from the steel of end A	630	2252
K1-BAT-24	ceiling	from the steel of end A	1210	2252
K1-BAT-25	ceiling	from the steel of end A	1789	2252
K1-BAT-26	ceiling	from the steel of end A	2369	2252
K1-BAT-27	ceiling	from the steel of end A	2949	2252
K1-BAT-28	ceiling	from the steel of end A	3529	2252
K1-BAT-29	ceiling	from the steel of end A	4109	2252
K1-BAT-30	ceiling	from the steel of end A	4688	2252
K1-BAT-31	ceiling	from the steel of end A	5268	2252
K1-BAT-32	ceiling	from the steel of end A	5823	2252

S04: first fix, by the electrician and the plumber

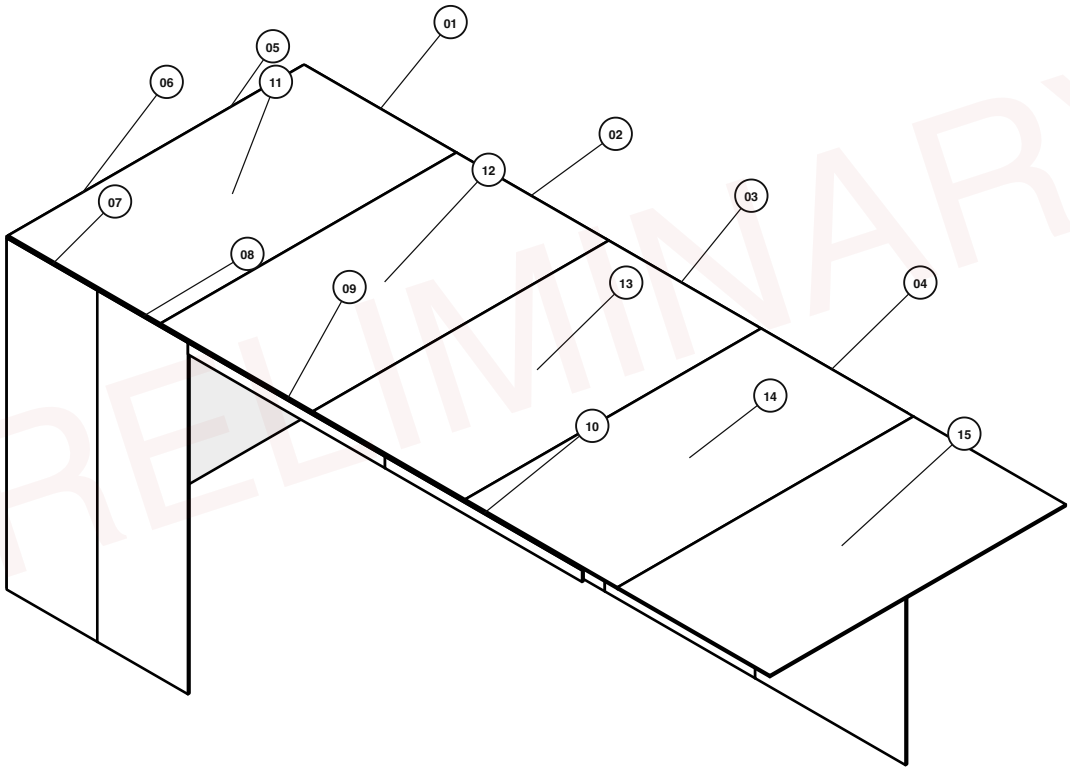
1. Keep the spaces between the battens clean and dry.
2. Help them mark the boxes and pipe routes from drawings E-101 and P-101 before the cables go in.
3. Do not start S05 until the 10 bar pressure test is written down and signed.

S05: spray foam, then the lining panels

05

Insulation and lining

Spray foam between the battens, let it cure and trim it flush; fix the



! HOLD: the S04 pressure test has passed. No welding or grinding on the shell once the foam

screw
600x F

Farm container studios	farm.databayt.org/docs/studio
Spray foam between the battens, let it cure and trim it flush;	**AS-05**
Scale NTS · Rev A2 · 2026-10-28 · mm	PRELIMINARY

PRELIMINARY

AS-05 Spray foam and lining panels · open full size

1. The spray-foam contractor fills between the battens with closed-cell foam, about 38 mm, walls and ceiling: 34.9 m² per studio. Mask the sliding-door frame, the window and the floor first.
2. After 24 hours, trim the foam flush with the battens.
3. Fix the 15 numbered lining panels (plywood 12, ids K1-01 to K1-15), ceiling first, then the walls. Every panel edge lands on a batten centre.
4. Screw with wood screws 4 × 30 every 200 mm along the edges and every 300 mm across the middle, about 600 per studio.
5. Check each cut-out against its box before fixing the panel; then check that no foam shows and the panel faces are in one plane.

S06 to S10: the bathroom (kit K2)

1. S06: stand the partition (70 thick, at 4578 from the finished end-A wall) and the cross wall, and fix their skins.
2. S07: fit the door frame and hang the bathroom door (600 wide, opening into the bathroom).
3. S08, the tiler: 12 mm cement board and a waterproof membrane on the wet walls, then tiles 300 × 600 from the floor up; grey safety vinyl in one sheet on the bathroom floor, coved 100 mm up the walls.
4. S09: set the shower tray (1000 × 900), its ledge and the fixed glass screen.
5. S10: hang the wall-hung WC, the 30 L water-heater box (450 × 450 × 300) on end wall B above the WC, the basin and the mirror cabinet in the wash lobby.

S11 to S14: oak planks, then the kitchen (kit K3)

1. S11: lay the oak planks (11.1 m² with 8% spare) over the living area from end A to the partition: rows along the container, end joints at least 300 mm from the row before.
2. S11, then: stand the fridge bay (550), the hob base (600) and the sink base (450) on their legs along the partition: one run of 1600, 600 deep.
3. S12: assemble the overhead cabinet and hang it on the steel rail, two people.
4. S13: scribe and fix the worktop 38 (620 deep), drop in the sink and the hob, fix the plinth.
5. S14: fix the hood shelf under the overhead cabinet and hang the hood.
6. Check: the walkway from the worktop to the sofa end is at least 650 (designed 658).

S15 to S18: bed, bedside table, seating, trims

1. S15, the bed (kit K4): fix the headboard to the back wall, bolt the frame, screw down the 20 slats, slide in the two storage boxes. The bed runs from end A to 1640.

2. S16, the bedside table (kit K5): glue the leg strips, screw the top and the shelf; it stands at 1640 to 1980.
3. S17, seating (kit K6): place the sofa (2000 to 3300) and slot together the table and the two chairs.
4. S18, trims (kit K7): scribe and pin the trims, caulk, paint off-white satin, fit the insect screens.

The workshop: cutting the kits

Only after revision B, from the survey.

1. Buy the boards for both studios from one batch. Measure ten sheets: plywood 18 must be 18.0 ± 0.3 mm, or tell the designer before cutting.
2. Rip into strips (± 0.5 mm), then crosscut; mark each part's id on a hidden face at once. Large panels: diagonals equal within 1 mm.
3. Edge-band what the cut list marks: L1 is one long edge, L2S2 is all round.
4. Drill with the templates: print T-01 at 100% and check its 100 mm bar first.
5. Dry-fit studio 1's kitchen and bed in the workshop before packing.
6. Pack one bundle per kit per studio, at most about 25 kg, labelled "Studio n / Kit / bundle x of y".

Per studio: 133 parts, listed in the cut list, with cutting diagrams C-101 to C-107 and the CNC files beside it.

What to buy, per studio

Item	Quantity
Plywood 12, sheets 2440 × 1220, floor	5
Plywood 12, sheets 2440 × 1220, linings	15
Plywood 18, sheets 2440 × 1220	8
Marine plywood 18, sheets 2440 × 1220 (bathroom)	3
Worktop blank 38, 3050 × 650	1
Treated battens 38 × 50, 2.4 m long	32, plus 10% spare
Polythene membrane 250 µm	15.3 m²
Oak planks 5 with underlay	11.1 m²

Item	Quantity
Spray foam, by the contractor	34.9 m²
Winged screws 5.5 × 50 (bag L)	192
Wood screws 4 × 30 (bag F), floor and linings	189 + 600
PU foam, 750 ml cans	12
ABS edge band	161 m

The full list with prices is the bill of materials.

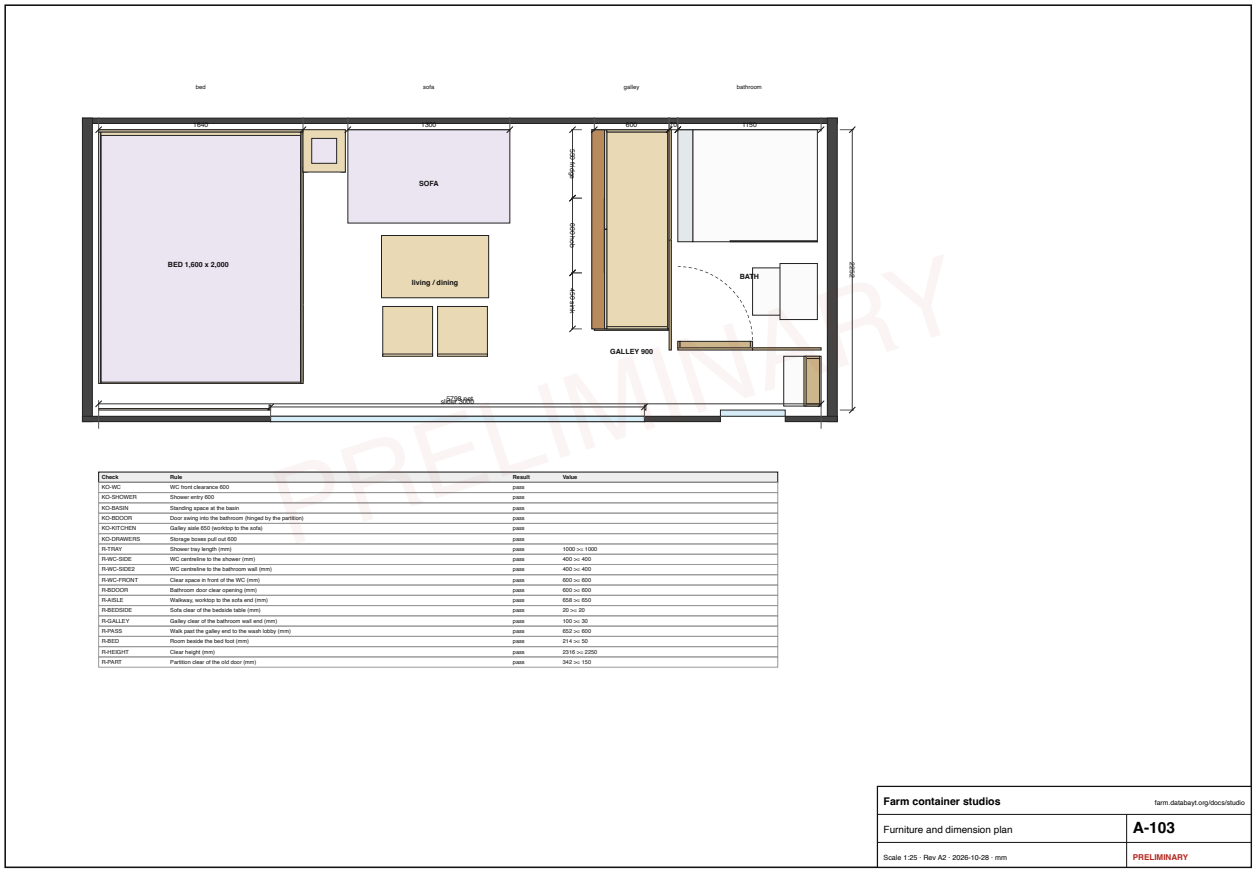
Before you call it finished

1. Doors and drawers run freely, the bathroom door swings clear, the sliding door locks, the insect mesh is tight.
2. No foam shows, every panel joint sits on a batten, no screw head stands proud.
3. Clear spaces: walkway at the sofa end at least 650, 600 in front of the WC, 600 at the shower entry.
4. The design itself passes its 33 checks.

Questions to answer before the step named

1. Which two containers are U1 and U2? Before Day 1 ends.
2. Is a welder booked for S01? Before Day 1 ends.
3. Who sprays the foam? No spray-foam contractor in Kigali is confirmed yet. Before S05.
4. Floor sheet thickness: 12 mm, unless the survey finds the old floor more than 5 mm out over 2 m; then 18 mm and a new revision. Before S02.

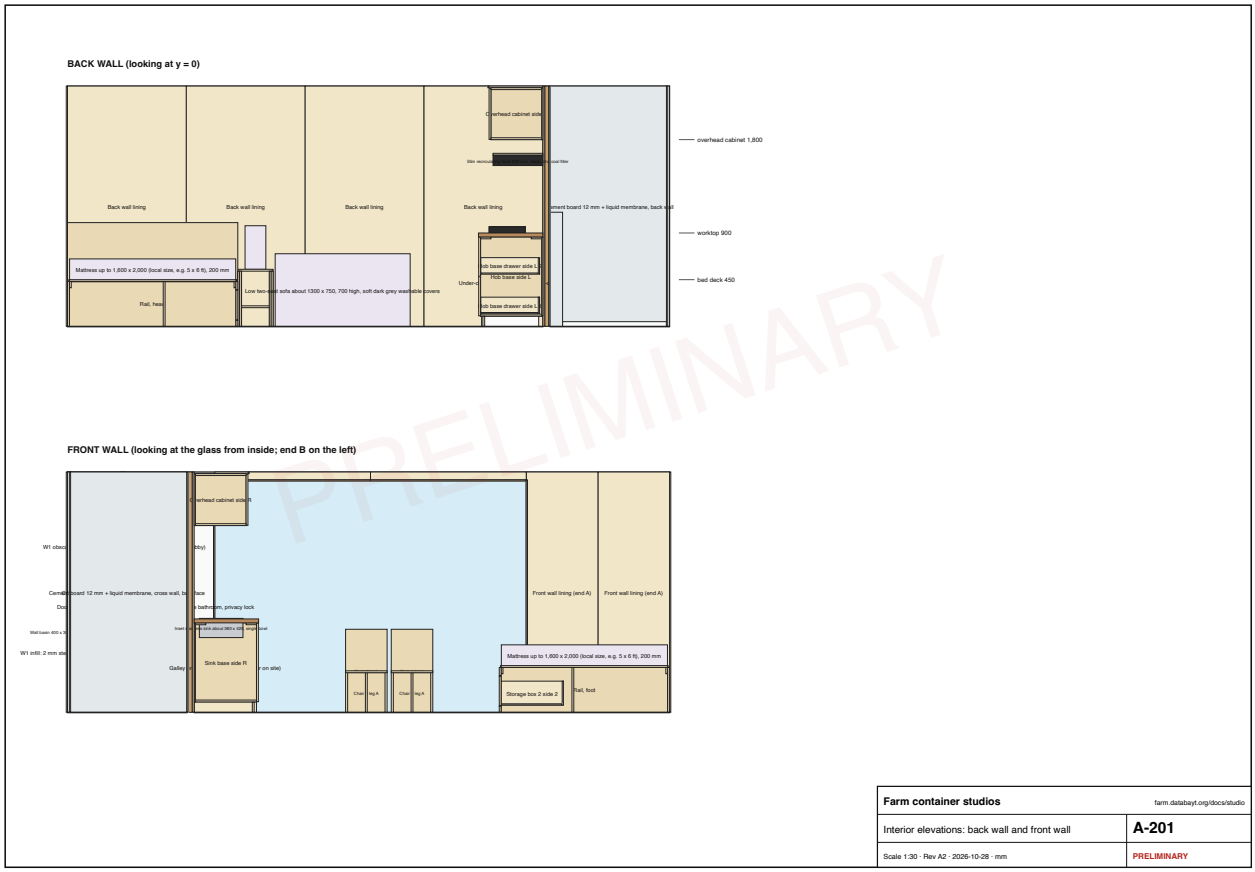
Drawings



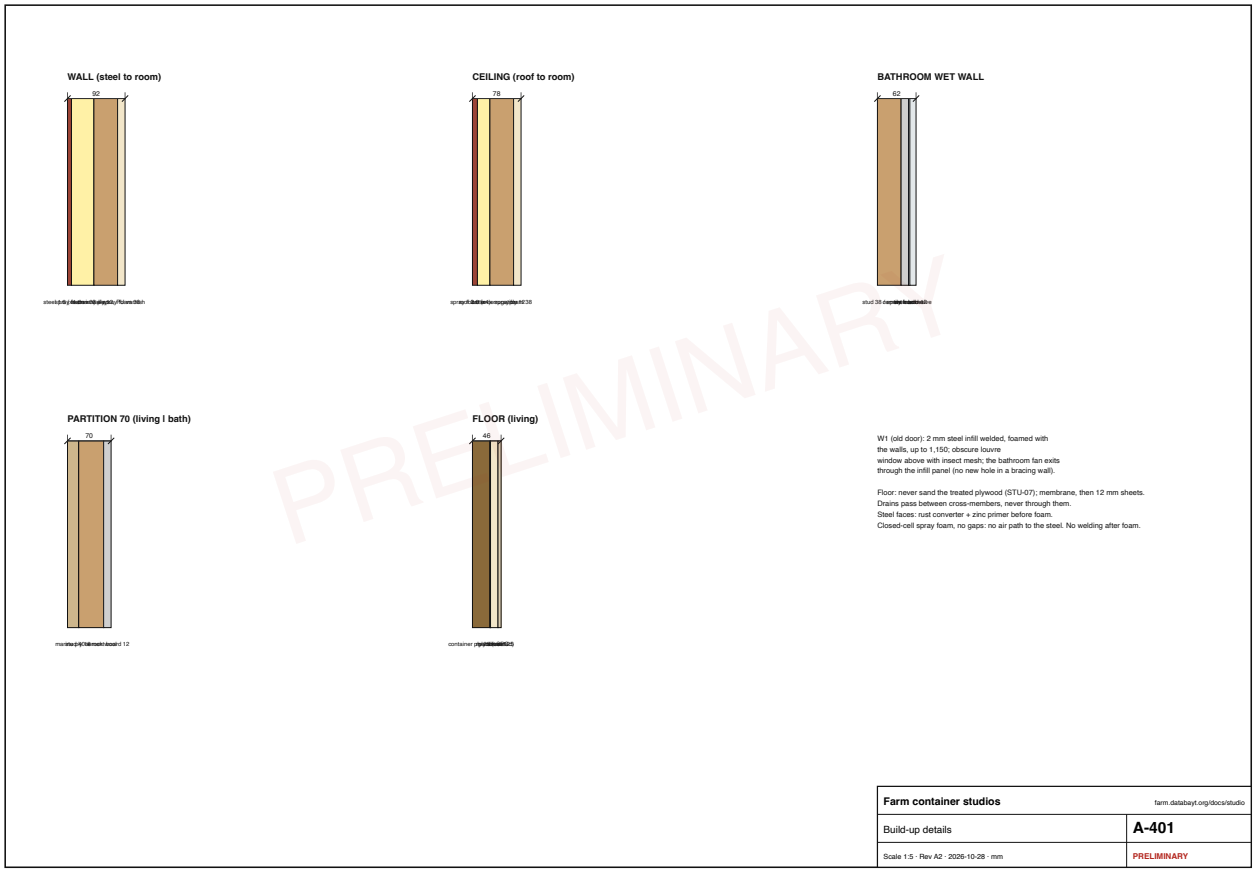
Check	Rule	Result	Value
KO-WC	WC front clearance 600	pass	
KO-SHOWER	Shower entry 600	pass	
KO-BASIN	Standing space at the basin	pass	
KO-BDOOR	Door swing into the bathroom (hinged by the partition)	pass	
KO-KITCHEN	Galley aisle 650 (worktop to the sofa)	pass	
KO-DRAWERS	Storage boxes pull out 600	pass	
R-TRAY	Shower tray length (mm)	pass	1000 >= 1000
R-WC-SIDE	WC centreline to the shower (mm)	pass	400 >= 400
R-WC-SIDE2	WC centreline to the bathroom wall (mm)	pass	400 >= 400
R-WC-FRONT	Clear space in front of the WC (mm)	pass	600 >= 600
R-BDOOR	Bathroom door clear opening (mm)	pass	600 >= 600
R-AISLE	Walkway, worktop to the sofa end (mm)	pass	658 >= 650
R-BEDSIDE	Sofa clear of the bedside table (mm)	pass	20 >= 20
R-GALLEY	Galley clear of the bathroom wall end (mm)	pass	100 >= 30
R-PASS	Walk past the galley end to the wash lobby (mm)	pass	652 >= 600
R-BED	Room beside the bed foot (mm)	pass	214 >= 50
R-HEIGHT	Clear height (mm)	pass	2316 >= 2250
R-PART	Partition clear of the old door (mm)	pass	342 >= 150

Farm container studios	farm.databayt.org/docs/studio
Furniture and dimension plan	A-103
Scale 1:25 · Rev A2 · 2026-10-28 · mm	PRELIMINARY

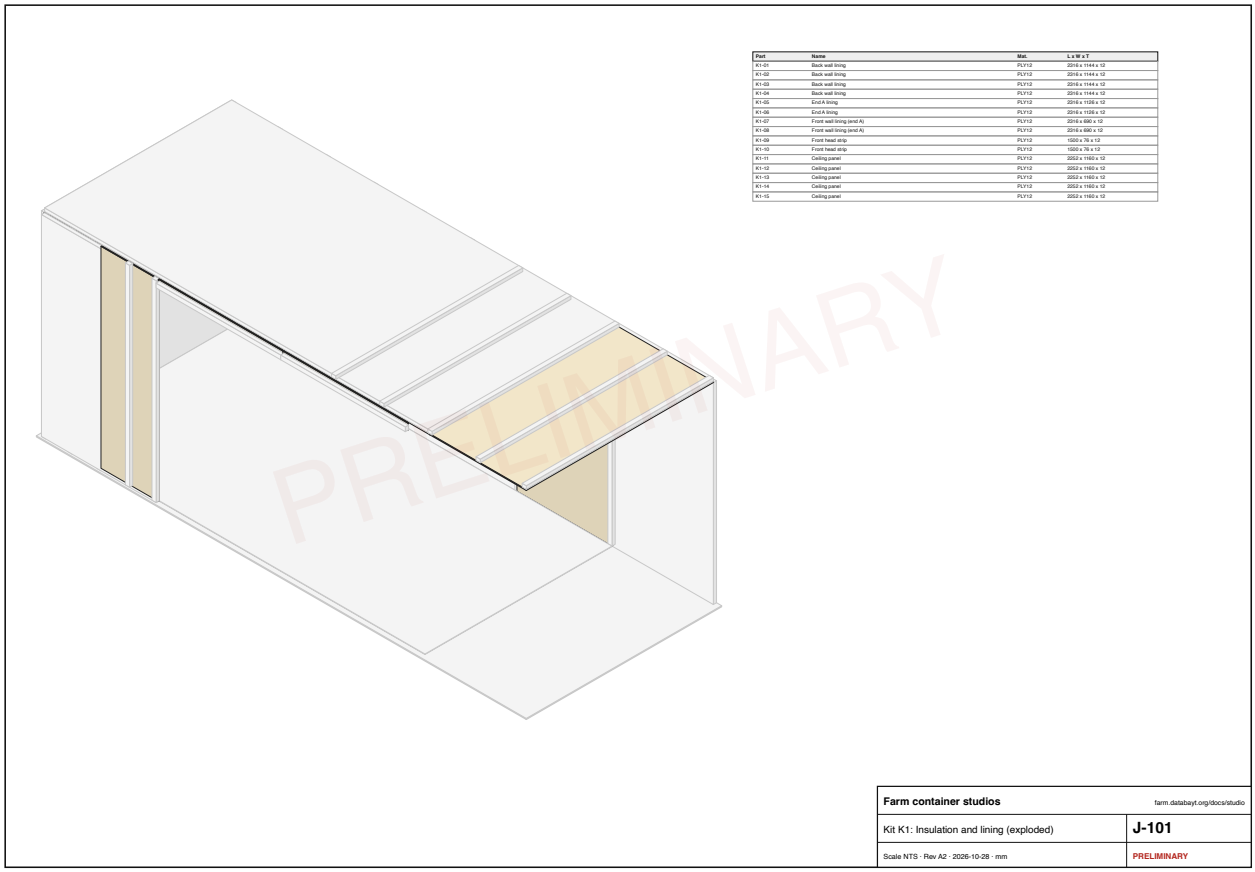
A-103 Furniture and dimension plan · open full size



A-201 Interior elevations: back wall and front wall · open full size



A-401 Build-up details · open full size



J-101 Kit K1: insulation and lining · open full size

Every other sheet, AS-01 to AS-27 and C-101 to C-107, is in the drawing list.